

Homology modeling and docking studies of *Comamonas testosteroni* B-356 biphenyl-2,3-dioxygenase involved in degradation of polychlorinated biphenyls

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ABSTRACT

Biphenyl dioxygenase is a microbial enzyme which catalyzes the stereospecific dioxygenation of aromatic rings of biphenyl congeners leading to their degradation. Hence, it has attracted the attention of researchers due to its ability to oxidize chlorinated biphenyls, which are one of the serious environmental contaminants. In the present study, the three-dimensional model of α -subunit of biphenyl dioxygenase (BphA) from *Comamonas testosteroni* B-356 has been constructed. The resulting model was further validated and used for docking studies with a class of chlorinated biphenyls such as biphenyl, 3,3'-dichlorobiphenyl and 4,4'-dichlorobiphenyl. The kinetic parameters of these biphenyl compounds were well matched with the docking results in terms of conformational and distance constraints. The binding properties of these biphenyl compounds along with identification of critical active site residues could be used for further site-directed mutagenesis experiments in order to identify their role in activity and substrate specificity, ultimately leading to improved mutants for degradation of these toxic compounds.

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1. Introduction

The microbial catabolic activities responsible for the degradation of aromatic compounds constitute an important aspect of the global carbon cycle. These activities are of considerable practical interest due to their potential to destroy toxic and persistent pollutants, a strategy known as bioremediation [1]. In the case of highly chlorinated, structurally diverse xenobiotics such as polychlorinated biphenyls (PCBs), the development of a practical bioremediation technology has been limited in part by the failure of existing microbial catabolic activities to effectively degrade these compounds [1,2]. Such failure may arise because these activities have not yet evolved to degrade compounds that have only recently been introduced into the biosphere. An important aspect of the adaptation of catabolic activities for bioremediation is the study of the structure and function of key catabolic enzymes. Such studies provide insight into the molecular basis of an important biological process, thereby facilitating the modification of enzyme specificity and the design of novel metabolic pathways. Aromatic oxytating dioxygenases are of considerable interest as they are potentially capable of initiating the degradation of numerous aromatic pollutants. Recent investigations provided evidence that these enzymes can be engineered to further increase their catalytic activity toward selected targeted pollutants, such as

polychlorinated biphenyls (PCBs) and polyaromatic hydrocarbons [3–7].

Biphenyl-2,3-dioxygenase (BPDO) catalyzes the initial reaction in the aerobic degradation of biphenyl and some PCBs. BPDO is a typical aromatic ring-hydroxylating dioxygenase, utilizing O_2 and electrons originating from NADH to transform biphenyl to cis-(2R,3S)-dihydroxy-1-phenylcyclohexa-4,6-diene [8–11]. This dihydroxylation prepares the ring for subsequent degradation by ring cleavage enzymes. The enzyme comprises three components which are: the terminal oxygenase, an iron-sulfur protein (ISPBPH) made up of an α -subunit (Mr 551,000) and a β -subunit (Mr 522,000), encoded by bphA and bphE, respectively; a ferredoxin (FERBPH, Mr 512,000) encoded by bphF; and a ferredoxin reductase (REDBPH, Mr 543,000) encoded by bphG. FERBPH and REDBPH were found to be involved in electron transfer from NADH to ISPBPH [8]. The Rieske center of the oxygenase component is then believed to receive the electron and pass it to a mononuclear Fe^{2+} which activates molecular oxygen for insertion into the substrate [11,12]. Coordination geometry of Fe^{2+} ion with conserved active site residues and O_2 is a major determinant, which affect the performance of BPDO for polychlorinated biphenyl degradation. The C-terminal portion of BphA is involved in substrate specificity and region specificity toward substrate analogs [13,14].

Despite the high degree of similarity among the BphA genes, substrate specificity differs drastically. For example, the BphA protein for *Burkholderia cepacia* LB400 and *Pseudomonas pseudoalcaligenes* KF707 differs in only 20 positions, and yet the biphenyl dioxygenase from LB400 preferentially dioxygenates

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ortho-substituted PCBs whereas the biphenyl dioxygenase from KF707 preferentially attacks para-substituted PCBs [15,16], indicating that the relatively few amino acid exchanges are responsible for the different substrate specificities. Because different substrate specificities have been observed for a relatively similar population of biphenyl dioxygenases, it is easy to envision that biphenyl dioxygenase can be modified in vitro for extended congener specificity. Detailed study on *B. xenovorans* LB400 BphA showed that amino acids T335, F336, N337, N338, I339, R340, I341, A377, F378, F384, G255, I258 and A268 [7,17–20], were found to influence the catalytic properties of the dioxygenase. Furthermore, structural analysis of *Rhodococcus* sp. strain RHA1 BphA1A2 (corresponding to BphAE of LB400) [21] and comparison with other aryl hydroxylating dioxygenases has confirmed the fact that some of the residues listed above are inside the substrate catalytic pocket and thus likely to interact with it to determine the regio specificity. Identification of the amino acid residues that are critical to determine the substrate and regio specificity and their mechanism of action will help design strategies to engineer newer, high performing enzymes. Several investigations using site-directed mutagenesis, and molecular evolution approaches have allowed the identification of amino acids that are critical for the enzyme specificity and region specificity toward chlorobiphenyls [13,14]. However, in the absence of detailed structural information, identifying all the amino acids responsible for congener specificity and those that might give rise to extended specificity remains an overwhelming challenge.

In the present study, the homology model of α -subunit of BPDO from *Comamonas testosteroni* B-356 has been constructed in order to get an in depth idea of its structural and functional aspects. The constructed model was further validated by structure analysis tools like PROSA and PROCHECK to analyze its structural integrity. After validating the consistency of the model it was further preceded for docking studies with PCB congeners (Chlorobenzenes). The docking studies were performed to understand the various important and conserved active site residues and interaction patterns that contribute to its ability to degrade polychlorinated biphenyls along with factors responsible for substrate specificity. Further, mutating the structurally conserved and functionally important active site residues could be used as potential targets for site directed mutagenesis leading to novel mutants with enhanced activity.

2. Methods

2.1. Software and hardware

Automated homology modeling by SWISS-MODEL [22], PHYRE [23], Geno3D [24] and MODELLER 7v7 (<http://salilab.org>). Model evaluation by PROCHECK [25], PROSA [26] and Verify-3D (http://nihserver.mbi.ucla.edu/Verify_3D/). Docking studies with version 4.0 of the AutoDock (<http://autodock.scripps.edu/>). Interactive visualization and analysis of molecular structures and related data, sequence alignments, docking results and conformational ensembles using UCSF Chimera [27].

2.2. Sequence alignment and homology modeling

The 3D model of α -subunit of BphA from *C. testosteroni* B-356 was built by homology modeling based on high-resolution crystal structures of homologous proteins. The protein sequence of α -subunit of BphA was obtained from Swiss-Prot (Q46372). The NCBI Basic Local Alignment Search Tool (BLAST), for the sequence similarities was used for searching the crystal structures of the closest homologues available in the Brookhaven Protein Data Bank (PDB). The results yielded by NCBI BLAST revealed that Cumene dioxygenase (CumA1A2) from *Pseudomonas fluo-*

rescens IP01 (PDB ID: 1wql) with a resolution of 2.2 Å as a suitable template and with an identity score of 76%. The coordinates of crystal structure of Cumene dioxygenase (CumA1A2) from *P. fluorescens* IP01 (PDB: 1wqlA) were used as template to build the model by pair-wise sequence alignment using clustalX software (<ftp://ftp.ebi.ac.uk/pub/software/clustalw2>). Homology modeling for α -subunit of BphA was performed by four automated homology modeling programs, SWISS-MODEL, PHYRE, Geno3D and MODELLER [22–24,28] using single as well as multiple template. The following templates were downloaded from the Brookhaven PDB (<http://www.rcsb.org/pdb/>): 1WQL, 1ULI and 2GBW. The backbone RMSD between model constructed using multiple templates and model constructed by best suitable template was determined using the structure comparison mode of chimera program.

2.3. Validation of model

The validation for structure models obtained from the different software tools (SWISS-MODEL, PHYRE, Geno3D and MODELLER) was performed by inspection of the Psi/Phi Ramachandran plot obtained from PROCHECK analysis [25]. Out of these, the model, constructed by MODELLER was finally chosen for further investigations as it possessed the best geometry and energy profile. The PROSA [26] test was applied for final model to check for energy criteria in comparison with the potential of mean force derived from a large set of known protein structures. Further, the compatibility of the model with its sequence was measured by Verify-3D (http://nihserver.mbi.ucla.edu/Verify_3D/). The root mean square deviation (RMSD) between the main chain atom of model and template was calculated by structural superimposition of template (1wql) and predicted structure of biphenyl dioxygenase for the reliability of the model using UCSF Chimera.

2.4. Docking studies of PCB congeners

Docking of α -subunit of BphA with the CBs was carried out with version 4.0 of the program AutoDock (<http://autodock.scripps.edu/>). This program combines a rapid energy evaluation through pre-calculated grids of affinity potentials with a variety of search algorithms to find suitable binding positions for a ligand on a given protein. The program allows torsional flexibility in the ligand. A standard protocol with an initial population of 50 randomly placed individuals, a maximum of 1.5×10^6 energy evaluations, a crossover rate of 0.80, and the elitism value of 1 were applied [29]. Proportional selection was used, where the average of the worst energy was calculated over a window of the previous 10 generations. For the local search, the so-called pseudo-Solis and Wets algorithm was applied using a maximum of 300 iterations per local search. The probability of performing local search on an individual in the population was 0.6, and the maximum number of consecutive successes or failures before doubling or halving the local search step size was 4. Fifty independent docking runs were carried out for each ligand. The Tripos force field with Kollman united atom charges for the protein, Gasteiger-Marsili charges for the ligand, and a distance-dependent dielectric constant was used for calculating the docking energy.

3. Results and discussion

3.1. Homology modeling of biphenyl dioxygenase α -subunit (BphA) from *C. testosteroni* B-356

Homology modeling is usually the method of choice when a clear relationship of homology between the sequence of target protein and at least one known structure is found. This approach

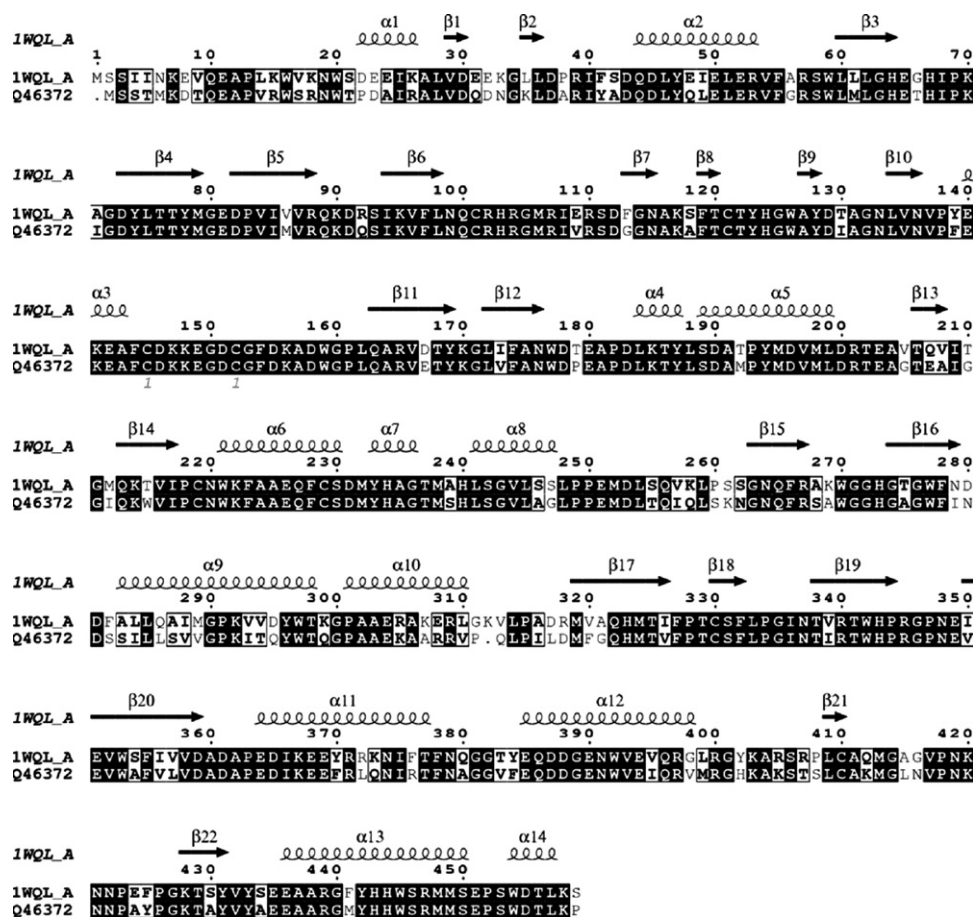


Fig. 1. Sequence alignment of BphA (Q46372) with CumA1A2 (1WQL). Conserved regions are represented by black boxes. The secondary structure for CumA1A2 is demonstrated with arrows for β -sheet and spiral for α -helices.

would give reasonable results based on the assumption that the tertiary structures of two proteins will be similar if their sequences are related [30]. The absence of the three-dimensional structure for *C. testosteroni* biphenyl dioxygenase (BPDO) in PDB prompted us to construct its homology model. BPDO is a major determinant of the PCB-catabolizing capabilities of biphenyl degrading strains, and the enzymes from different strains possess significantly different congener transforming abilities. Crystal structures of dioxygenases from different bacterial species have already been determined and available in PDB [31]. Among the dioxygenases with known three-dimensional structures, the Cumene dioxygenase (PDB ID: 1wql) showed the highest sequence identity with BphA from *C. testosteroni* B-356. Primary sequence alignment showed that *C. testosteroni* B-356 BphA and α -subunit of Cumene dioxygenase from *P. fluorescens* IP01 (PDB ID: 1wql) share 76% identity (similarity 88%). At this level of sequence identity, it is good enough to use crystallographic structure of CumA1A2 from *P. fluorescens* IP01 as a template, in order to obtain high quality alignment for structure prediction by homology modeling. The final alignment of BphA (Q46372) and CumA1A2 (PDB ID: 1wqlA) with secondary structural elements of CumA1A2 is shown in Fig. 1.

The three-dimensional structure provides valuable insight into molecular function and also enables the analyses of its interactions with suitable chlorobiphenyls (CBs). Four models were constructed by different protein prediction tools: Geno3D, Swiss-Model, Phyre and Modeller. The predicted models were checked for psi and phi torsion angles using the Ramchandran plots. A comparison of the results obtained from the different software tools, shows that one of the model (MODEL.A) generated by Modeller is more acceptable in

comparison to those generated by Geno3D, Phyre and Swiss-Model (Table 1).

Like Cumene dioxygenase crystal structure, the BphA homology model is predominantly α -helical in nature (38% α -helical) with only 18% β -sheeted residues. The structure was primarily organized in 14 long α -helices and 16 short β -sheets as shown in Fig. 2.

3.2. Validation of the predicted structure

The final chosen model (MODEL.A) shows higher internal consistency and reliability as compared to others. Its PROCHECK analysis based on the inspection of Psi/Phi Ramchandran plot showed 91.1% of the residues in the core region, 7.9% residues in the allowed regions, 1.0% in the generously allowed regions and no residues in disallowed region as shown in Fig. 3. The G-factors, indicating the quality of covalent and bond angle distance, were 0.04° for dihedrals, -0.20° for covalent, and overall -0.05° . The overall main-chain and side-chain parameters, as evaluated by PROCHECK,

Table 1

Comparative analysis of the validated models from Modeller, Geno3D, Phyre and Swiss-Model.

Model	Program	Ramachandran plot quality (%)		
		Allowed	General	Disallowed
Model A	MODELLER	99.0	1.0	0.0
Model B	SWISS-MODEL	98.9	0.8	0.3
Model C	PHYRE	98.9	0.8	0.3
Model D	Geno3D	96.8	2.4	0.8
1WQL	–	98.6	1.1	0.3



Fig. 2. The 3D structure of BphA. The α -helix is represented by magenta spirals, β -sheet by green arrows.

are all very favorable. The Ramachandran plot characteristic and G-factors confirm the quality of predicted model.

In order to investigate whether the interaction energy of each residue with the remainder of the protein is negative, a second test was done to apply energy criteria using Prosa II energy plot. The Prosa analysis of the model showed maximum residues to have negative interaction energy with very few residues displayed positive interaction energy as shown in Fig. 4. The overall interaction energy of the model was -8.78 kcal/mol, which is quite similar to the template CumA1A2 (-8.65 kcal/mol).

Next, the MODEL.A was subsequently checked by VERIFY-3D graph as shown in Fig. 5. The compatibility score above zero in the VERIFY-3D graph corresponded to acceptable side chain environments.

Homology modeling for *C. testosteroni* B-356 BphA was also performed using multiple templates instead of single template.

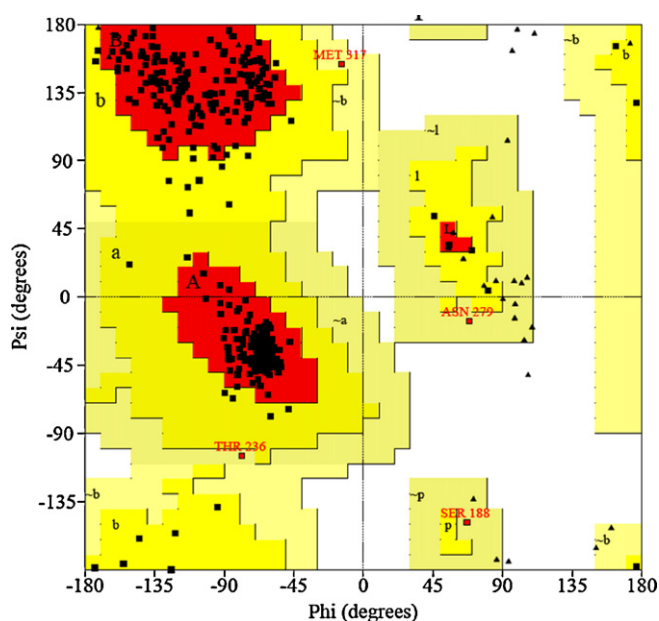


Fig. 3. Ramachandran plot of BphA model built using NCBI Structural Analysis and Verification Server.

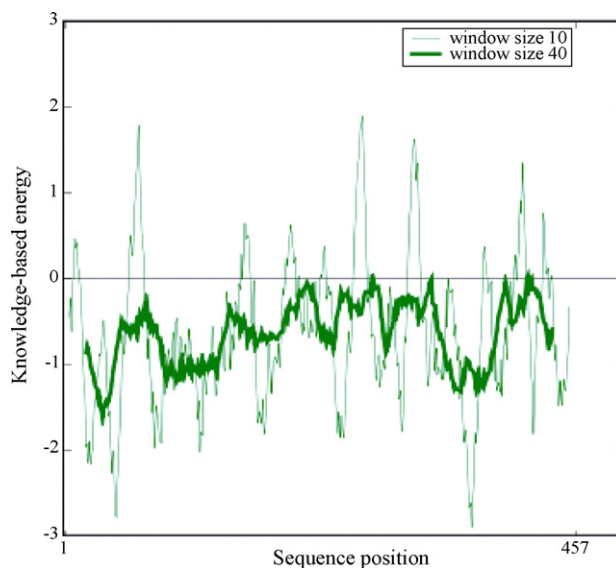


Fig. 4. Prosa energy plot calculated for *Comamonas testosteroni* B-356 BphA model.

The backbone RMSD between single template:target and multiple template:target models (0.5 Å) suggest that both models either constructed by multiple template or single template are good enough for further studies. Further, the superimposition between the template (purple) and the model (cyan) is shown in Fig. 6. The backbone RMSD 0.220, between the final *C. testosteroni* B-356 BphA model and the template CumA1A2 from *P. fluorescens* IP01 crystal structure support that generated model is reasonably good and quite similar to template. Hence, the final model which proved to be well validated in terms of geometry and energy profiles suggests that the model is good enough to be a starting point for our next phase of docking studies.

3.3. Active site of *C. testosteroni* B-356 BphA

The sequences and structures of BphA and CumA1A2 are conserved due to their similar biological functions. Like CumA1A2, the BphA active site is composed of two domains, the catalytic domain (residues 1–58 and 176–457) and the Rieske domain (residues 59–166). The active site pocket, which is lined mostly by hydrophobic residues (Phe227, Trp390, Phe 376, Asp386, Phe382 and Ala234) as shown in Fig. 7, is located between the core β -sheet and α -helices in the catalytic domain of the α -subunit where the helices $\alpha 7$, $\alpha 8$, $\alpha 9$, $\alpha 10$, $\alpha 11$, $\alpha 12$ sheet $\beta 13$, loop $\alpha 11$ – $\beta 12$ and loop $\beta 13$ – $\beta 14$ form the substrate binding pocket. Comparative active site analysis of *C. testosteroni* B-356 BphA and CumA1A2 shows var-

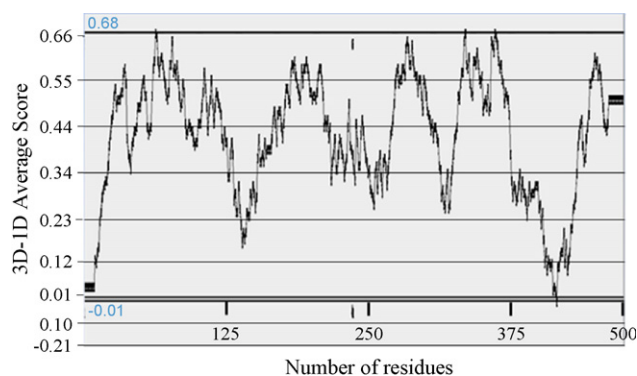


Fig. 5. The 3D profiles verified results of BphA model, where residues with positive compatibility score are reasonably folded.

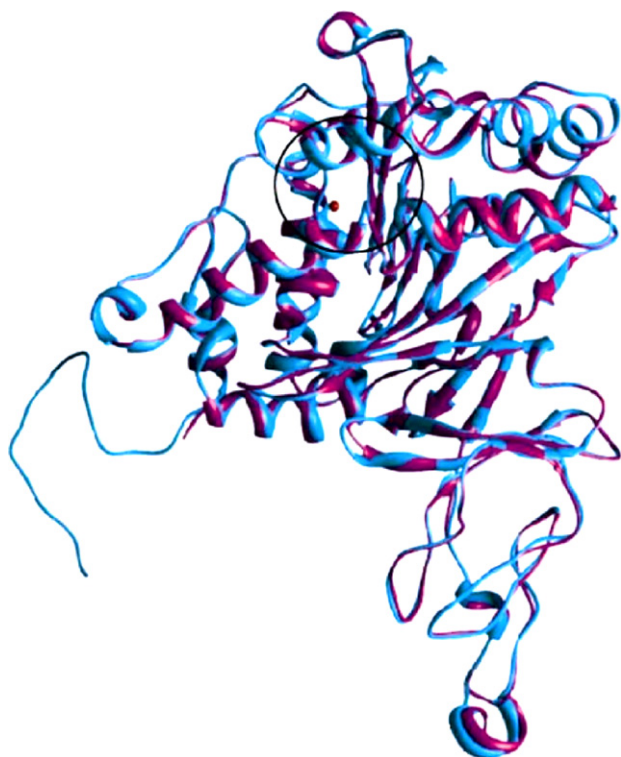


Fig. 6. Superimposed backbones of BphA model (cyan) and the template CumA1A2 (purple). Active site residues have been encircled. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

ious characteristic differences in the respective active site residues. For example, residues like Gly319, Phe318, Ile283, Ser286, Val287, Phe382 and Ser238 in BphA are replaced by Ala321, Val320, Leu284, Ala287, Ile288, Tyr384 and Ala239 in CumA1A2, respectively. These relative differences in the active site should undoubtedly affect the conformation and presumably play a significant role in the substrate specificity.

Superimposition of catalytic site of related enzymes shows that the residues lining the substrate binding pocket are well conserved among them as shown in Fig. 8. Alignment result shows that *C. testosteroni* B-356 BphA active site is more similar to related enzymes from *P. fluorescens* IP01 and *Rhodococcus* sp. Strain RHA1 as compared to active site of *Sphingobium yanoikuyae* B1. The Rieske domain, which is composed of 10 β -strands and one α -helix and considered to be involved in electron transfer, adopts a similar structure as that of the ferredoxin component of a BDO system [32]. The Rieske domain contains a [2Fe–2S] Rieske cluster, which is coordinated by His102, His123, Cys100, and Cys120, all of which are completely conserved among the related enzymes.

3.4. Docking of PCB congeners

Finally, docking study of the model was performed in order to elucidate its structural and functional relevance in terms of substrate binding and specificity. PCB congeners were successfully

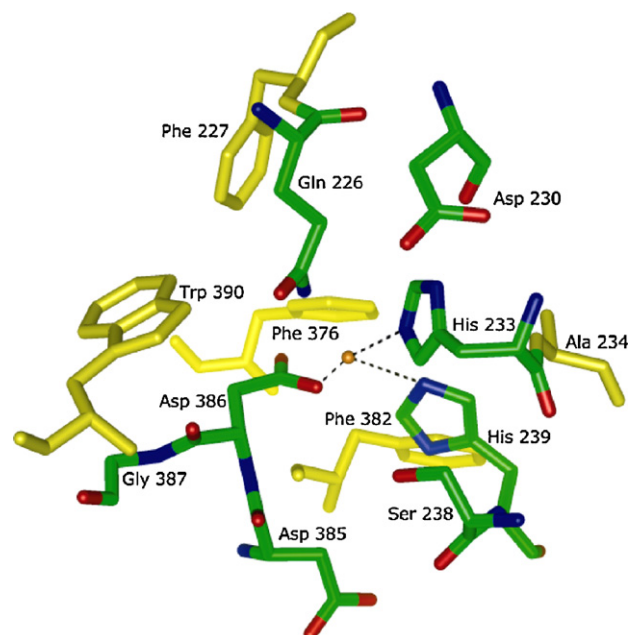


Fig. 7. Stick representation displaying active site residues (atom color) located around the Fe^{2+} ion. Hydrophobic residues are shown in yellow color. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

docked onto the active site of biphenyl-2,3-dioxygenase, according to the mentioned docking protocol. Fig. 9a shows docking of biphenyl into the active site of the BphA. A careful inspection of the binding pocket indicated that biphenyl adopted a position in a hydrophobic cage surrounded by Phe227, Trp390, Phe376, Ala234 and Phe382. Dock score was calculated as -5.91 kcal/mol and distance of Fe ion from C2 position of proximal phenyl ring was 4.64 Å and from C3 position, 5.07 Å. Fig. 9b shows 3,3-dichlorobiphenyl docking into the active site of the BphA. The phenyl ring of 3,3-dichlorobiphenyl was oriented a little further away compared to biphenyl. Dock score was calculated as -6.11 kcal/mol and distance of Fe ion from free C2 position of proximal phenyl ring was 5.75 Å and from C3 position, it was 5.49 Å. These results hence are pointing toward the importance of distance constraints in the binding site of BphA, where the closer the distance of the C2 and C3 positions of the binding compound with the Fe ion, more will be the oxygen utilization which will ultimately lead to potent catalysis. Finally, Fig. 9 shows the docked position of 4,4-dichlorobiphenyl in the active site of BphA. In this case, the phenyl ring was oriented in close proximity to residues involved in coordination geometry with Fe ion. Dock score was determined as -4.75 kcal/mol and distance of Fe ion from C2 position of proximal phenyl ring was 2.39 Å and from C3 position, it was 3.27 Å. Since, here the C2 and C3 positions of the phenyl ring are comparatively closer, hence it can be assumed that 4,4-dichlorobiphenyl might be more catalytically potent as compared to the other two.

Table 2 shows the results of the docking experiments in terms of calculated free energy of binding, corresponding experimental K_m values for oxygen utilization [33] and distance of Fe ion from

D386	F227	O226	H233	D230	M231	H321	A234	H239	O320	G319	F318	I283	L284	S286	V287	I334	F382	F376	L331	W390	G387	D385	S238	MODEL A
D388	F228	O227	H234	D231	M232	H322	A235	H240	O321	A321	V320	L284	L285	A287	I288	I336	Y384	F378	L333	W392	G389	D387	A239	1WOL_A
D378	F218	O217	H224	D221	M222	L309	A225	H230	O312	A311	M310	L274	L275	A277	I287	I326	F374	F368	L323	W382	G379	D377	S229	1U L I_A
D360	F201	N200	H207	D204	W210	N295	A213	H212	H293	G292	—	K269	F270	G272	D273	T308	L356	F350	L305	M364	G361	D359	S217	2GBW_A
*	*		*	*		*		*										*	*		*	*		

Fig. 8. Amino acid residues at active site of biphenyl-2,3-dioxygenases from *C. testosteroni* B-356, *Pseudomonas fluorescens* IP01(1wql), *Rhodococcus* sp. strain RHA1(1uli) and *Sphingobium yanoikuyae* B1(2gbw). The essential residues in catalytic domain which form coordinate geometry with mononuclear iron (Fe ion) are boxed. Other conserved residues are represented by asterisk.

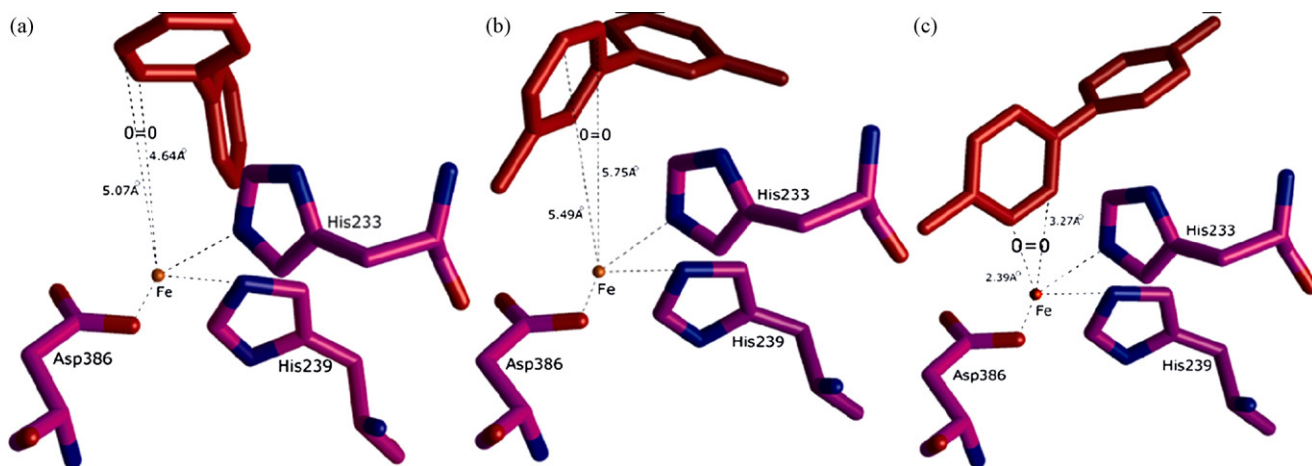


Fig. 9. Docked conformation of (a) biphenyl, (b) 3,3-dichlorobiphenyl, and (c) 4,4-dichlorobiphenyl into the active site of BphA (atom color). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

Table 2

Coordination geometry of Fe-ion along with the oxygen and different PCB congeners in each docking experiment along with their dock score.

S. no.	Compounds	O ₂ utilization (μM) [33], <i>K_m</i> (O ₂)	Distance of Fe ion from PCB congeners (Å)	
			C2-Fe	C3-Fe
1.	Biphenyl	28	4.64	5.07
2.	3,3'-Dichlorobiphenyl	190	5.75	5.49
3.	4,4'-Dichlorobiphenyl ^a	NR	2.39	3.27

NR: not reported.

^a In presence of 4,4'-dichlorobiphenyl, the initial rate of O₂ uptake increased linearly with the concentration of O₂ [33].

biphenyl congeners. The latter being an important determinant in the oxidation of PCB congeners, was used as an important factor for analyzing the docking results. It was observed that the mononuclear iron is coordinated by His233, His239 and Asp386 (completely conserved among the relative enzymes) and with oxygen responsible for oxygenation of substrates.

The docking studies with PCB congeners revealed that their binding mode with BPDO were different with respect to each other, which correlated reasonably well with experimental values of O₂ utilization, *K_m*(O₂) [33]. In case of the docked conformation of 3,3'-dichlorobiphenyl, the higher *K_m* for O₂ utilization tendency could be well correlated with its little farther distance from the Fe ion. While, a comparatively closer conformation of biphenyl with the Fe ion is in lieu with its O₂ utilization property. Finally, in 4,4-dichlorobiphenyl, the phenyl ring was oriented in more close proximity to the Fe ion along with other residues involved in coordi-

nation geometry with Fe ion, suggesting that 4,4-dichlorobiphenyl can be a better substrate for BPDO as compared to other PCB congeners. However, further experimental validation is still needed to confirm the assumed theory. PCB congeners are placed into the active site of BPDO in approximately similar positions but the distance of Fe ion with the respective C3 and C4 is different, which is decisive in the ultimate oxidation of PCB congeners. Hence, it can be said that the distance of Fe ion to PCB substrates is inversely proportional to oxygen utilization.

Further, all the three proteins (2GBX, 1ULJ and BphA model) were superimposed with their respective biphenyl bound and unbound structures, to determine the changes in the side chain conformations of active site residues. The active site residues nearest to Fe ion and in close proximity to biphenyl did not exhibit significant shift, on the other hand, some flexibility was observed in the residues (1ULJ: Met310, Leu275, Asn273, Leu274, Ile278,

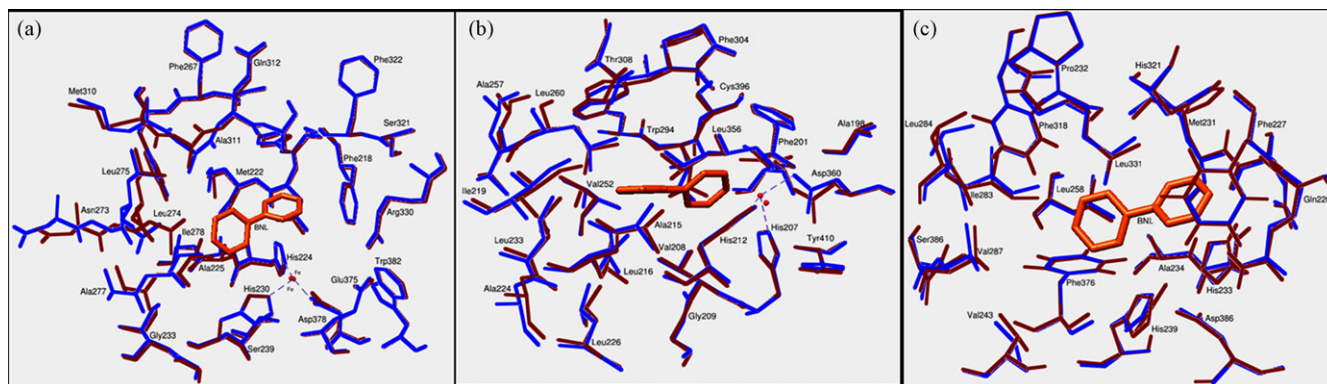


Fig. 10. Superimposition of active site amino acid residues in their biphenyl bound (blue) and biphenyl unbound (red) conformations. (a) *Rhodococcus* sp. strain RHA1; 1uli, (b) *Sphingomonas yanoikuyae* B1; 2gbw, and (c) *C. testosteroni* B-356 model. Energy computations were done with the GROMOS96 implementation of Swiss-PdbViewer. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

Ala277 and Ala311; 2GBX: Ala257, Leu260, Val252, Ile219, Ala215, Ala224, Ala224 and Leu233; BphA: Leu284, Ile283, Val287, Val243 and Leu331) which are closer to distal ring of biphenyl. This flexibility may help to accommodate different substrates of various sizes and shapes. The possible reason for negligible changes in conformation closer to Fe ion might be that protein active sites are large enough and presence of bulky hydrophobic residues which tend to retain their conformations. The results have been further verified by minimizing the docked structures, in order to check the stability of protein–ligand complex (Fig. 10).

4. Conclusion

In the present work the 3D model of BphA was constructed in order to accomplish its molecular modeling and docking studies. The model was validated and further used for docking analysis with some well known PCBs. The resulting docking solutions were analyzed for binding pattern and conformational analysis. It was observed that the proximity of the C2 and C3 position of the docked PCBs were determinant of their oxygen utilization and ultimate catalysis. Hence, this work could be further useful for the mutagenesis studies where mutations created at various positions of the binding site may alter the substrate specificity and potency. Further work is still going on in our laboratory for the same.

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